

§4.4.

L'Hospital's Rule

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \rightarrow \text{type } \frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

$$\Rightarrow \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Remark: $\Rightarrow$ L'Hospital's rule can be applied repeatedly when feasible.

2) $\lim_{x \rightarrow a} f \cdot g \rightarrow \text{type } 0 \cdot \infty$

$$\text{then } f \cdot g = \frac{f}{\frac{1}{g}} \rightarrow \frac{0}{0}$$

$$f \cdot g = \frac{g}{\frac{1}{f}} \rightarrow \frac{\infty}{\infty}$$

3) $\lim_{x \rightarrow a} [f(x)]^{g(x)} \rightarrow \text{type } 0^0, \text{ or } \infty^0, \text{ or } 1^\infty$

$$\text{Let } y = [f(x)]^{g(x)}, \text{ then } \ln y = g(x) \ln f(x).$$

$$\Rightarrow \lim_{x \rightarrow a} [f(x)]^{g(x)} = e^{\lim_{x \rightarrow a} \ln y} = e^{\lim_{x \rightarrow a} g(x) \ln f(x)} \rightarrow \text{apply L'Hospital's Rule.}$$

§4.5.

Curve Sketching P 307 - 313.

§4.7.

Solving optimization Problem.

$$\boxed{\max/\min_x Q(x)}$$

1. Understand the problem.

2. Draw a diagram.

3. Introduce notation.

4. Set up equations and express Q in terms of some symbols.

5. Transform the Q equation into a function of one variable if not by using some other relations.

6. Solve the max/min problem.

First Derivative Test for Absolute Extreme Values P 324.

§4.9

Antiderivatives

$$f(x) \rightarrow F(x)$$

such that

$$F'(x) = f(x)$$

Chapter 5.

§5.1.

upper and lower sums for approximating areas

§5.2.

Definition of a Definite Integral

f , $a \leq x \leq b$, $\Delta X = (b-a)/n$, $\{x_0=a, x_1, x_2, \dots, x_{n-1}, x_n=b\}$.

$x_i^* \in [x_{i-1}, x_i]$, $i=1, 2, \dots, n$ any sample points

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta X.$$

$\xrightarrow{\text{upper limit}}$ $\xrightarrow{\text{integrand}}$ $\xrightarrow{\text{lower limit}}$ $\xrightarrow{\text{net area}}$ Riemann Sum.

more simply,

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta X.$$

Properties:

$$\int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$\int_a^a f(x) dx = 0$$

$$\int c dx = c(b-a)$$

$$\int c f(x) dx = c \int f(x) dx$$

$$\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$$

$$\int_a^b f(x) dx \geq 0 \quad \text{if } f(x) \geq 0 \text{ in } [a, b]$$

$$\int_a^b f(x) dx \geq \int_a^b g(x) dx \quad \text{if } f(x) \geq g(x) \text{ in } [a, b]$$

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a) \quad \text{if } m \leq f(x) \leq M \text{ in } [a, b]$$

§5.3

Fundamental Theorem of Calculus

Part I: Let $g(x) = \int_a^x f(t) dt$, $a \leq x \leq b$, then

$$g'(x) = f(x).$$

part II: $\int_a^b f(x) dx = F(x) \Big|_a^b$
 $= F(b) - F(a)$

where F is any antiderivative of f .

§5.4.

$$\int f(x) dx = F(x) \text{ means } F'(x) = f(x)$$

Table

$$\int x^r dx = \frac{x^{r+1}}{r+1} + C \quad (r \neq -1)$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

$$\int e^x dx = e^x + C$$

$$\int a^x = \frac{a^x}{\ln a} + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$\int \frac{1}{x^2+1} dx = \arctan x + C$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$$

$$\int \sinh x dx = \cosh x + C$$

$$\int \cosh x dx = \sinh x + C$$

$v(t)$ - velocity of a particle moving along a line

$$\text{displacement} = \int_{t_1}^{t_2} v(t) dt$$

from $t=t_1$ to $t=t_2$

$$\text{distance traveled} = \int_{t_1}^{t_2} |v(t)| dt$$

from t_1 to t_2

§5.6.

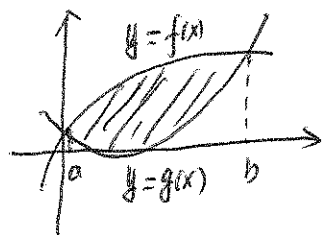
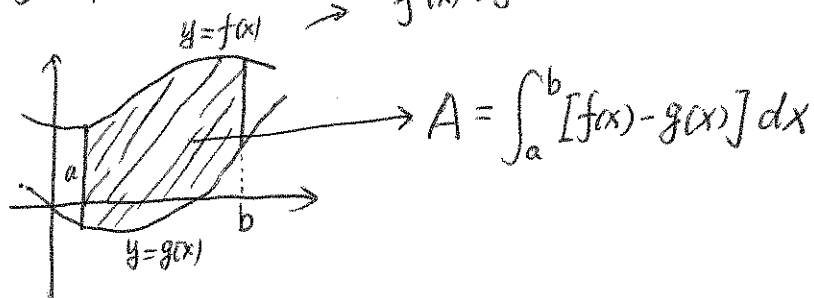
Substitution Rule: $u = g(x)$ then

$$\int f(g(x)) g'(x) dx = \int f(u) du$$

$$\int_a^b f(g(x)) g'(x) dx = \int_{u(a)}^{u(b)} f(u) du$$

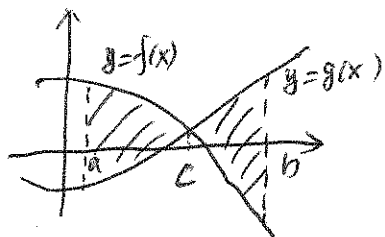
Chapter 6

§6.1. $f(x) \geq g(x)$ on $[a, b]$



$\begin{cases} y=f(x) \\ y=g(x) \end{cases} \Rightarrow x=a \text{ or } b, \text{ then}$

$$A = \int_a^b [f(x) - g(x)] dx$$

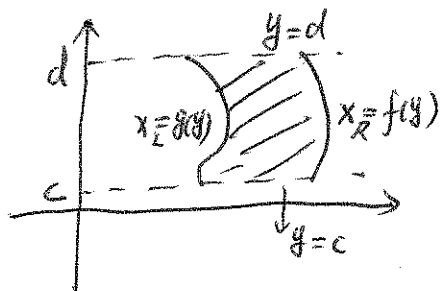


$$A = \int_a^b |f(x) - g(x)| dx$$

$$= \int_a^c [f(x) - g(x)] dx + \int_c^b [g(x) - f(x)] dx$$

$\downarrow$
 $f(x) \geq g(x)$ on $[a, c]$

$\downarrow$
 $g(x) \geq f(x)$ on $[c, b]$



$$A = \int_c^d (x_R - x_L) dy$$

$$= \int_c^d [f(y) - g(y)] dy$$